

Leveraging Machine Learning Models for Seismic Prediction: A Study on Model Performance and Optimization

Rahul Singh Pawar

Department of Information Technology
Medicaps University

Indore(M.P), India

rahuls.pawar@medicaps.ac.in

https://orcid.org/0009-0002-7915-9518

Prashant Panse

Department of Information Technology
Medicaps University

Indore(M.P), India

prashant.panse@medicaps.ac.in

https://orcid.org/0000-0001-8129-1214

Jyoti Kukade

Department of Information Technology
Medicaps University

Indore(M.P), India

jyoti.kukade@medicaps.ac.in

Soumya Neema

Department of Information Technology
Medicaps University

Indore(M.P), India

soumyaneema360@gmail.com

Gagan Singh

Department of Information Technology
Medicaps University

Indore(M.P), India

singh.gagan2003.as@gmail.com

Harsh Goyal

Department of Information Technology
Medicaps University

Indore (M.P), India

yg599172@gmail.com

Abstract—Predicting earthquakes has been a persistent challenge in seismology, attracting significant interest from the researchers. Accurate earthquake prediction is crucial for mitigating seismic risks and enhancing early warning systems. This study investigates the application of advanced machine learning algorithms—XGBoost, Stacking Regressor, Support Vector Machine (SVM), K-Nearest Neighbors (K-NN), Long Short-Term Memory (LSTM), and Random Forest to predict earthquake magnitudes in the Indian subcontinent. The fulfilment of these machine learning models is systematically evaluated and compared using historical seismic data to identify the most effective approach. Among the models tested, XGBoost showed significant improvements after hyperparameter optimization, with accuracy increasing from 92.76% to 92.81%, alongside reductions in MAE (0.3215), MSE (0.1895), R^2 (0.1548) and RMSE (0.4353). Model evaluation is conducted using key performance metrics, including the Mean Squared Error (MSE), coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) ensuring robust and reliable predictions. This study aids in enhancing seismic hazard evaluation by utilizing data-driven methods.

Keywords—Earthquake Prediction, LSTM, Machine Learning, Seismic Data, Stacking Regressor, XGBoost, Long Short-Term Memory, K-Nearest Neighbour, Random Forest, Support Vector Machine.

I. INTRODUCTION

Earthquakes are among the most destructive natural catastrophes, and they have serious implications for human populations and critical infrastructure. The quest to attain accurate seismic predictions has been a priority in the scientific research agenda for decades. Traditional earthquake prediction methods are largely dependent on seismological data and past trends. The methods are not highly accurate and lack the ability to provide early warnings, hence minimizing their capability in providing timely warnings[1].

Natural disasters result in huge loss of human life and properties, but precautionary measures and safety regulations can reduce their effects. While some scientists are of the opinion that earthquakes cannot be predicted, others are optimistic about predictive methods based on variations in

electric and magnetic fields, ionospheric activity, animal behaviour, and past experience[2]. A successful earthquake prediction model should be able to effectively evaluate the location, size, and timing of seismic activity. As of now, there is no complete mechanism for such predictions. A successful system would allow authorities to deploy resources for evacuation and shut down critical infrastructure, such as nuclear power plants and power grids, thus saving lives and property. This research employs laboratory micro-earthquake simulation input parameters to track seismic energy buildup in fault zones to predict upcoming earthquake magnitudes. Understanding the correlation between seismic activity and geophysical data is crucial for further research in earthquake prediction[3]. Advanced machine learning techniques have been employed to identify earthquake precursors and predict the opportunity for future seismicity by examination. Apart from this, such systems may provide early alarm systems capable of warning societies of a prospective threat and give them enough time to take preventive steps[4].

Machine learning (ML) has been a successful substitute for traditional methods over the last decade. Methods related to ML, including gradient boosting, deep learning, and neural networks, are very effective in detecting patterns in large datasets that may be missed by traditional models[5]. Long Short-Term Memory (LSTM) network has gained massive popularity because it can be analyse structured as well as sequential data, hence being most suitable to forecast complex natural disasters, such as earthquakes[6].

The precise forecasting of earthquakes includes the application of a set of variables, including magnitude, location, and time. From previous research, these variables have extremely high correlation and uncertainty and hence need advanced computational methods for precise forecasting. Techniques for machine learning like XGBoost, uses gradient boosting to enhance forecasting capacity, and LSTM which can understand long-term correlations in time-series data, have grown increasingly popular in earthquake research. In addition, these algorithms can be combined with ensemble learning techniques like the Stacking Regressor, which uses a

combination of base models to improve predictive power[2] [5].

The application of sophisticated machine learning models is examined in this study—Support Vector Machine, XGBoost, K-Nearest Neighbors (K-NN), Long Short-Term Memory (LSTM), Stacking Regressor, and Random Forest for predicting earthquake magnitudes. The study compares the performance of these models to determine the most accurate approach. The XGBoost model is further optimized to enhance prediction accuracy. Predicting earthquake magnitudes remains a complex challenge, requiring robust analytical methods. The models are trained on historical seismic data and assessed using key evaluation to ensure reliable predictions[2] [5].

II. LITERATURE SURVEY

The prediction of accurate earthquakes is a important challenge due to complexity and unpredictability of seismic activities. Traditional earthquake prediction methods have relied on seismic sensors and historical patterns but these techniques have high errors and less accuracy. Thus because of this Machine learning models and deep learning techniques are used which works on a large amount of data sets which reduces the error and increases accuracy[7].

Forecasting earthquake magnitudes and locations [8] has compared three Machine learning model which are XG boost, Stacking, regressor and LSTM for earthquake prediction in Duzce, Türkiye and found that XG boost has given better result than other two Machine learning model in accuracy and reliability.

Study [2] has found that Neural Networks has provided the best predictive performance among XGboost, Random forest and SVM. In detecting earthquakes in California, U.S. With the help of Geospatial visualization we can analyse, interpret and visualize geographical and spatial data and thus this tool help in mapping and detecting earthquakes. By introducing folium which is used to create interactive maps and Geopandas that work with geographical shapes like points, lines and polygons helps researchers and disaster management teams to visualize affected areas in real time.

[9] This Paper proposes a uses small earthquake that has been occurred in predicting larger ones. They have used two different machine learning techniques to comprehend i.e. long short-term memory (LTSM) and Random Forest the patterns in the seismic data and according to the study they have found Random Forest is highly effective for predicting the earthquakes. This study has been done in Sichuan-Yunnan region using seismic datasets containing earthquakes from 1970-2021.

Research [10] used advanced deep learning methods, including as transformer models like Bi-directional Long Short-Term Memory (Bi-LSTM), Long Short-Term Memory (LSTM) to forecast earthquake magnitudes in a number of places, including Indonesia, Japan and the Hindu-Kush Karakoram (HKKH) region. Metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE) were used for evaluating the models, and the performance showed that the LSTM model performed best for the Japanese dataset.

[11] This paper proposes uses neural networks (NN) algorithm for predicting earthquakes. The dataset has been collected from 1900-2019 has been used in which 80% has

been used in training the model and 20% has been used in testing the model and major attribute which has been taken in the data is of ground vibration. They have found that Neural networks model has performed better than other machine learning algorithms such Random Forest, logistic regression, and support vector machine (SVM).

Study[12] has used three machine learning models which are Histogram based gradient boosting, adaptive boosting and soft voting classifier to improve prediction accuracy. Soft voting classifier combines two best performing models and achieved an accuracy of 94%.

Ensemble learning, i.e. combining multiple ML models can help us to improve our accuracy and deep learning algorithms such as Long short-term memory (LSTM) and Recurrent Neural networks (RNN) have shown good results in time-series seismic data although they require large datasets for optimal performance. Accuracy in earthquake prediction is greatly increased by using the machine learning techniques [11]. While no machine learning model has given 100% accuracy currently but with advancements in feature selection and data processing this percentage can improve with time.

III. METHDOLOGY

A. Data Collection

The dataset for earthquake prediction using machine learning algorithms is a valuable resource that contains a variety of seismic event data. It often includes earthquake magnitudes, depths, locations, and timestamps. Considering there are significant fault lines in this area, which is known for its tectonic significance, this data is crucial for evaluating the seismic activity and the threats related to earthquakes. The Indian Subcontinent Earthquake dataset was compiled from the USGS Earthquake Catalogue by Kaggle from November 2000 to November 2024, spanning a 24-year period and it can be available via the following URL: <https://www.kaggle.com/datasets/ankitd7752/indian-subcontinent-earthquake-data-2000-to-2024>.

The feature description for the dataset, which had 17,805 samples and 22 features at the time of training, is shown below[6]:-

- time: the time when the incident took place. Since the epoch, time are expressed in milliseconds.
- Latitude: Latitude in decimal degrees. Southern latitudes have negative values.
- Longitude: Longitude in decimal degrees. Western longitudes have negative values.
- depth: The event's depth measured in kilometres.
- mag: The event's magnitude.
- magType: The procedure or technique utilized to determine the desired magnitude
- nst: The overall count of seismic stations that are used to pinpoint the location of earthquakes.
- gap: The greatest azimuthal distance (in degrees) between stations that are adjacent to each other.
- dmin: The horizontal distance (in degrees) between the closest station and the epicenter.

- rms: Using all weights, the root-mean-square (RMS) duration of the residual, expressed in seconds.
- net: The ID of a contributor to the data source for the event that took place.
- id: An event's special identification number.
- types: A list of product categories related to this event, separated by commas.
- location: a specific geographic area close to the event.
- type: Seismic event type.
- locationSource: The network that first provided the event's reported location.
- magSource: the organization which first determined the event's reported magnitude.
- horizontalError: The degree of uncertainty regarding the event's reported position in kilometers.
- depthError: The depth error and its three main errors are highlighted by a vertical line.
- magError: The said magnitude of the event was inaccurate.
- magNst: The sum of all seismic stations at which the magnitude of an earthquake is measured.
- status: indicates if the occurrence has been examined by a human.

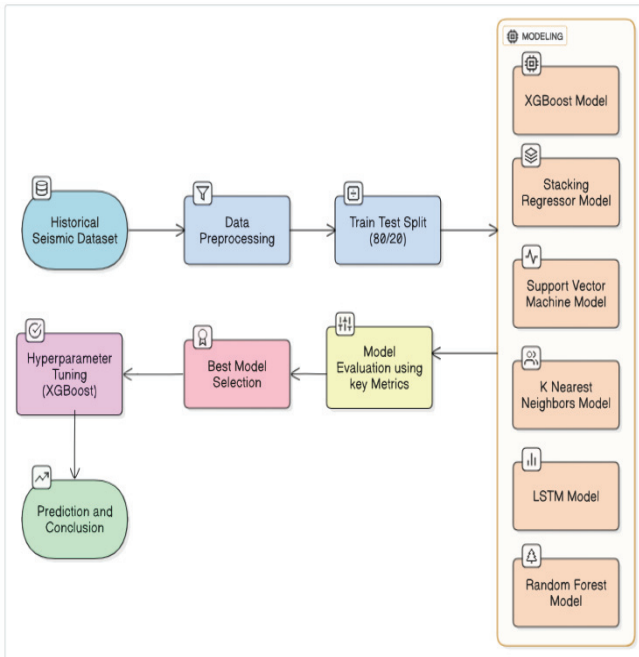


Fig. 1. Block diagram of the methodology for earthquake magnitude prediction using machine learning models.

At analysis each dataset went through a detailed preliminary step prior to the analysis. This was due in order to guarantee and maintain data consistency and integrity. Data was cleaned of its anomalies or interfering and the intention here was to keep the results accurate. Unit standardisation, dealing with missing values, preprocessing and normalization were the steps in preprocessing. The research is sought to improve the predicted accuracy of the machine learning

models used in the analysis by generating a clean and consistent dataset[8]. In conclusion, this meticulously compiled dataset serves as the foundation for earthquake magnitude prediction in the Indian Subcontinent Region. The machine learning models used in this study are supported and improved by historical data and efficient preprocessing approaches[8].

B. Machine learning models

In our study we have used six different machine learning techniques for earthquake prediction. Here our machine learning models are predicting on the basis of seismic data collected from Indian subcontinent. The main attributes taken for prediction are latitude, longitude and depth. The model used are XGBoost, Support Vector machine, Random Forest, stacking regressor, LSTM, and K-Nearest neighbours (KNN).

1) XGBoost (Extreme Gradient Boost)

XGBoost is a machine learning model which is a type of ensemble learning method, it combines multiple models or trees iteratively to improve performance by using a technique known as Gradient Boosting. XGBoost creates extremely accurate predictions by recursively fixing mistakes from earlier trees, which makes it perfect for handling complicated and complex, non-linear relationships in seismic data. Even with large-scale datasets, the technique is renowned for its amazing accuracy, quick computing speed, and economic memory usage. In light of its superior precision and ability to handle structured and tabular data, it is often employed in real-world applications.

XGBoost performed well in this investigation, with an initial accuracy of 92.76%, MAE of 0.3235, and R^2 score of 0.1415. However, the default parameter settings posed dangers of overfitting and inefficient learning. The default learning rate of 0.1 accelerated training but resulted in instability, and the number of estimators (100) was insufficient to capture complicated patterns. Furthermore, the uncontrolled max depth caused overfitting, and the lack of limits on min_child_weight and gamma allowed for unnecessary splits, lowering model efficiency. The use of subsample = 1.0 and colsample_bytree = 1.0 increased tree correlation, which reduced model diversity. Furthermore, the lack of explicit regularization inflame overfitting, and the lack of early stopping resulted in training that performed better than ideal.

The important and critical hyperparameters were optimized to increase stability and generalization. Decreasing the learning velocity to 0.07 allowed for more consistent convergence over 300 estimators while also enhancing the learning depth. The maximum depth was chosen at 6 in order to limit excessive tree growth and continue to be highly predictive. The efficiency of the model was improved by lowering unnecessary splits, adjusting gamma to 0.2, and setting min_child_weight to 3. The incorporation of randomization with subsample = 0.85 and colsample_bytree = 0.85 increased generalization and reduced tree correlation. By implementing explicit regularization with reg_lambda = 3 and reg_alpha = 1, overfitting was efficiently controlled. In conclusion, introducing an early termination with 20 rounds optimized training efficiency. With an accuracy of 92.81%, an MAE of 0.3215, an MSE of 0.1895, an RMSE of 0.4353, and an R^2 of 0.1548, the model's effectiveness and efficiency were improved following these modifications, indicating how

effectively these adjustments improved prediction accuracy. The result is summarized in Table I.

TABLE I. COMPARISON OF ACCURACY OF XGBOOST MODEL

Model	Baseline Accuracy	Optimized Accuracy
XGBoost	92.76%	92.81%

2) Random Forest

Random forest is an ensemble approach to learning that reduces the overfitting and enhancing accuracy by basing decisions on many different trees. It can work well with large amount of data and useful in capturing nonlinear relationships. First it creates multiple decision tree and then makes predictions using all trees and then combines the result for final prediction.

In our study, Random Forest has achieved an accuracy of 92.54% with an MAE of 0.3333 which we can say it performed well in comparison to other machine learning models.

3) Stacking Regressor

Stacking Regressor is a highly developed ensemble learning method which improves prediction by integrating many machine learning models. It doesn't depend on a single model for prediction. It works on two levels, first one is base model in which multiple machine learning models make decision independently and second one is meta model which take these predictions as input and generate final decision as output. It can handle complex data as well as reduces overfitting.

In our study, stacking regressor has achieved the highest accuracy of 92.78% and MAE of 0.3228, which makes it the best machine learning model.

4) Long Short-Term Memory (LSTM)

LSTM is a unique type of Recurrent Neural network (RNN) which handles data in sequential form and hold information for extended amount of time. It doesn't treat all inputs independently like old neural networks models, whereas it recognizes patterns across time dependent data. It uses foregate which decides which information should be discarded from past and input gate which decides what new information should be added and output gate which choose what information will be used for final prediction.

It has relatively low performance as compared to others machine learning models having R² score of 0.0730 and an accuracy of 92.35%.

5) K-Nearest Neighbors (KNN)

KNN is used for regression and classification tasks. It uses Euclidean distance to measure the distance between points. It predicts values based on k most similar data points and it is less effective in earthquake prediction due to complexity of seismic patterns. It is slow and noisy data can affect its performance and not accurate as advanced models like stacking regressor and XGBoost. Whereas it is easy understand and implement and no training is required in KNN.

$$\bar{y} = \frac{1}{k} \sum_{i=1}^k y_i$$

Let k be the number of nearest neighbors. Here, y_i is the actual value and $\bar{y}$ is the predicted value, which is the average of the values of the k nearest neighbors for the i^{th} data point.

In our study KNN has achieved accuracy of 92.44% with an MAE of 0.3379 showing moderate performance.

6) Support Vector Machine

SVM is used for both regression and classification tasks. SVM divides the data by the help of line in higher dimensions with the largest margin, wider the margin the better generalization of data and if data is not linearly separable then SVM transforms it into a higher dimension using kernels. It works slow for large datasets and choosing the right kernel can be difficult.

In our study, SVM has the accuracy of 91.59% making it the lowest among all the machine learning models and negative R² score of -0.2422 indicating poor generalization for earthquake prediction.

C. Performance Metric

The machine learning models underwent a thorough evaluation employing various quantitative metrics aimed at delivering an in-depth analysis of their prediction accuracy and dependability[5]. The Chosen Performance metric are: -

1) Mean Absolute Error (MEA):

The Mean Absolute Error is the average absolute difference between estimated and observed values within a dataset. This metric represents the arithmetic mean of the absolute errors, focusing exclusively on their magnitude without accounting for their direction. MAE provides important information regarding the global reliability of the predictions and helps in determining how well do the model predict earthquake events that were finally recorded. A lower MAE indicates improved model accuracy[8].

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}_i|$$

Let n be the total number of observations. Here, y_i is the actual value and $\bar{y}_i$ is the predicted value for the i^{th} data point. The term $|y_i - \bar{y}_i|$ represents the absolute error between them.

2) Mean Squared Error (MSE):

Mean Squared Error (MSE) is the average squared difference between actual and predicted values, effectively measuring the extent of deviation in predictions. This property makes it particularly useful in scenarios where minimizing substantial prediction errors is critical. Moreover, MSE helps to improve the model execution by indicating what can be improved and ultimately achieve more accurate prediction confidence[8].

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2$$

Let n be the total number of observations. Here, y_i is the actual value and $\bar{y}_i$ is the predicted value for the i^{th} data point.

3) Root Mean Squared Error (RMSE):

Root Mean Square Error (RMSE) is a statistical measure used in statistics and machine learning that evaluates the accuracy of a predictive model. The reason why it is preferred

to optimize the model, during training by model optimization algorithms due the efficiency and differentiability is performed. More specifically, a lower RMSE indicates a better model and higher RMSE are the opposite[8].

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2}$$

Here, y_i is the actual value and $\bar{y}_i$ is the predicted value for the i^{th} data point. Let n be the total number of observations.

4) R^2 (Coefficient of Determination):

The values ranging from 0 to 1, the R^2 value indicates the proportion of the target variable's variance that can be explained by the model. A model with higher R^2 showing the scope of model to find critical patterns where there is a high consistency between predicted data and actual data of the model is. This is a very important number when evaluating how well the model is reproducing the drift in the data as a whole[8].

Considering the use of these parameters, this study evaluates the accuracy of earthquake forecasts and identifies best prediction method. We want to develop a more precise and accurate predictive model for earthquake magnitude forecasting by utilizing machine learning models.

IV. RESULT AND DISCUSSION

Distinct metrics, such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), R^2 score, and Accuracy (%) were used to assesses the different machine learning models performed in earthquake prediction. Table II provides a summary of the results obtained: -

Among this models, the **Optimized XGBoost** demonstrated the best performance and efficiency, achieving the lowest error values and the highest R^2 and **accuracy (92.81%)**. The **XGBoost model** is improved by fine-tuning important hyperparameters. The **max depth** was set to **6** to avoid excessive tree growth, and the **learning rate** was lowered to **0.07** for stable convergence over **300 estimators**. **Subsample (0.85)** and **colsample_bytree (0.85)** decreased tree correlation for improved generalization, while **Gamma (0.2)** and **min_child_weight (3)** minimized needless splits. Overfitting was reduced by regularization (**reg_lambda = 3**, **reg_alpha = 1**), and efficiency was increased by **early stopping (20 rounds)**. With **92.81% accuracy**, **MAE of 0.3215**, **MSE of 0.1895**, **RMSE of 0.4353**, and **R2 of 0.1548**, these improvements enhanced performance.

TABLE II. EVALUATION OF DIFFERENT MACHINE LEARNING MODELS PERFORMANCE

Model	MAE	MSE	RMSE	R^2	Accuracy (%)
Optimized XGBoost	0.3215	0.1895	0.4353	0.1548	92.81
XGBoost	0.3235	0.1925	0.4387	0.1415	92.76
Random Forest	0.3333	0.2025	0.4501	0.0966	92.54
Stacking Regressor	0.3228	0.1908	0.4369	0.1488	92.78

LSTM	0.3420	0.2078	0.4559	0.0730	92.35
KNN	0.3379	0.2089	0.4571	0.0681	92.44
SVM	0.3759	0.2785	0.5278	-0.2422	91.59

The **Stacking Regressor** performed well, achieving a MAE of **0.3228**, MSE of **0.1908**, RMSE of **0.4369**, R^2 of **0.1488**, and an accuracy of **92.78%**. It closely followed the Optimized XGBoost in terms of accuracy and error reduction, making it a strong alternative.

The **Random Forest model** showed slightly higher error values, with an MAE of **0.3333** and an MSE of **0.2025**, leading to an accuracy of **92.54%**. The **LSTM model**, known for its effectiveness and performance in sequential data processing, achieved an accuracy of **92.35%**, with an RMSE of **0.4559** and an R^2 score of **0.0730**. Although, despite its suitability for time-series data, the low R^2 score indicates that LSTM study predictive ability was limited with static features as in the present study.

The **K-Nearest Neighbors (KNN) algorithm** performed similarly to LSTM, with an accuracy of **92.44%** but slightly higher error metrics. Conversely, **SVM exhibited the weakest performance**, with the highest error values (MSE: **0.2785**, RMSE: **0.5278**) and the lowest accuracy (**91.59%**). Additionally, the negative R^2 score of 0.2422 indicates that

SVM failed to model the data effectively, most likely because of its parameter tuning sensibility or the nonlinear relationship in seismic. Overall, the findings suggest that ensemble-based model XGBoost offer superior predictive accuracy for earthquake forecasting, making them promising candidates for real-world applications.

A. Visualizing the results

This section presents various data visualizations showing the achievements of the ML Models that we used in our exploration as follows: -

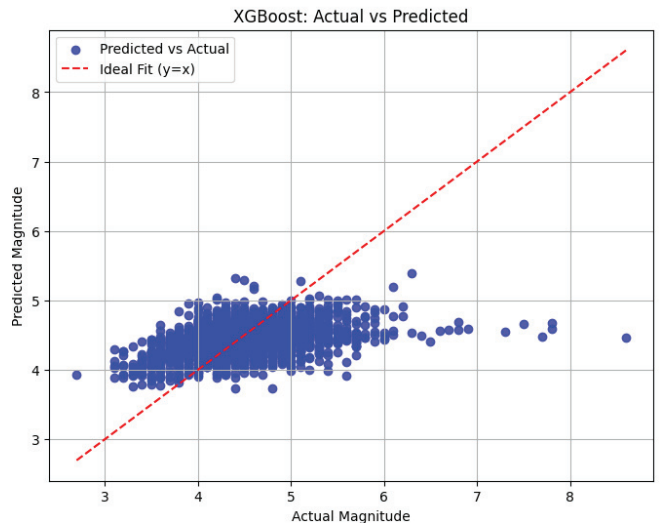


Fig. 2. Analysis of XGBoost Model: Predicted vs Actual Earthquake Magnitudes

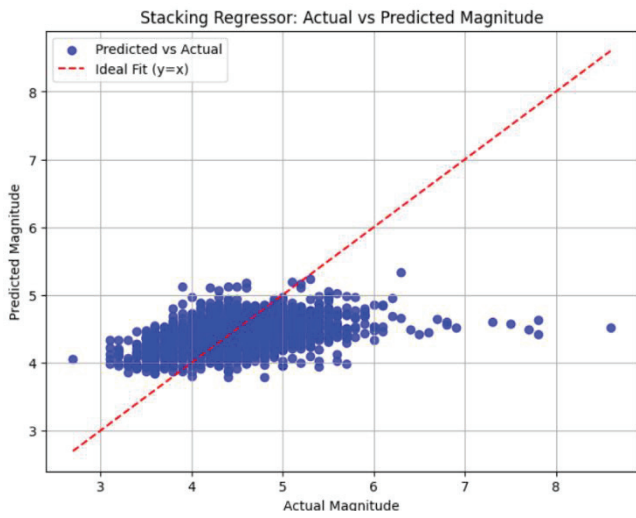


Fig. 3. Analysis of Stacking Regressor Model: Predicted vs Actual Earthquake Magnitudes

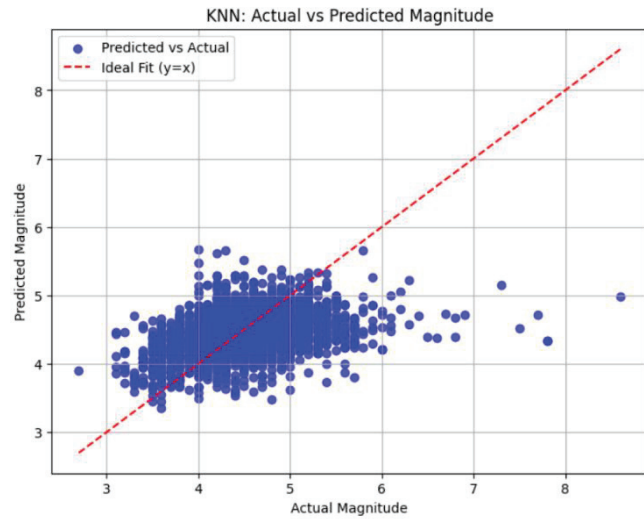


Fig. 4. Analysis of KNN Model: Predicted vs Actual Earthquake Magnitudes

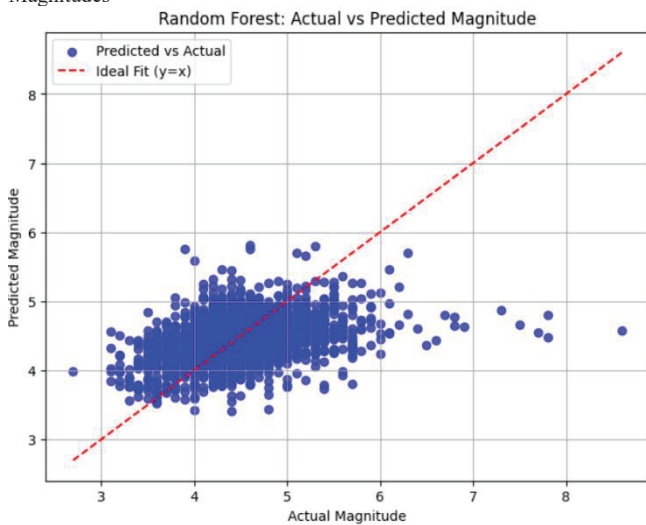


Fig. 5. Analysis of Random Forest Model: Predicted vs Actual Earthquake Magnitudes

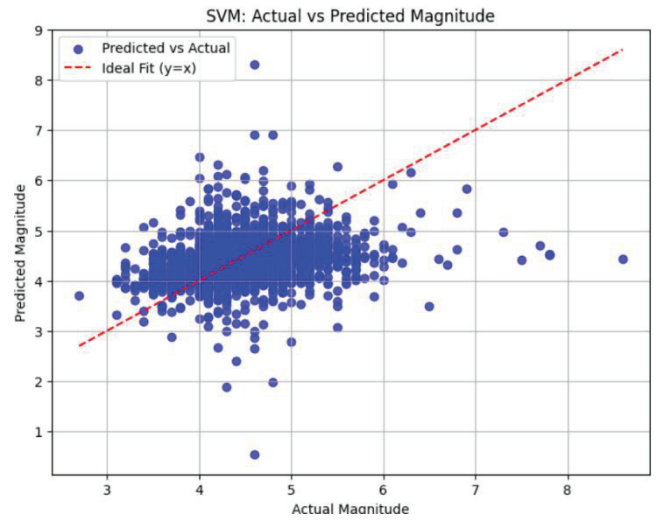


Fig. 6. Analysis of SVM Model: Predicted vs Actual Earthquake Magnitudes

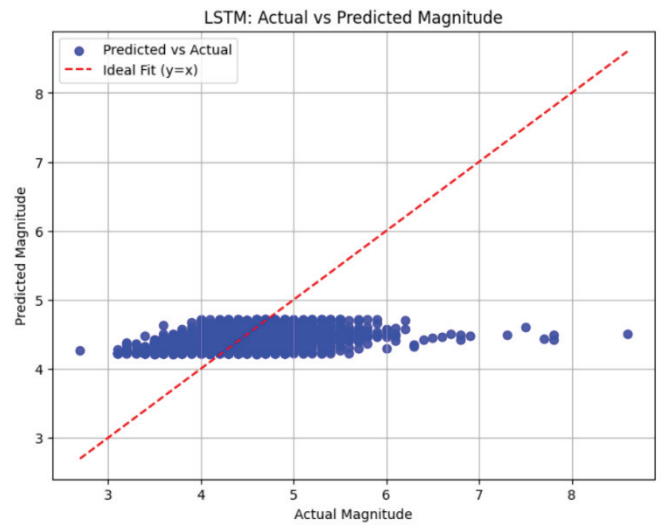


Fig. 7. Analysis of LSTM Model: Predicted vs Actual Earthquake Magnitudes

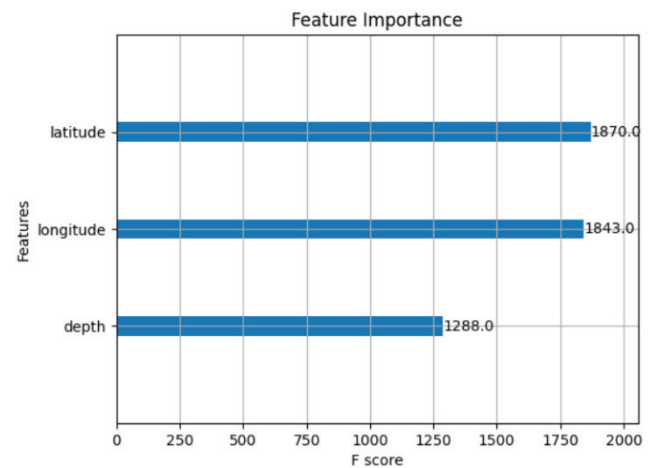


Fig. 8. Evaluating Feature Importance in Earthquake Magnitude Prediction

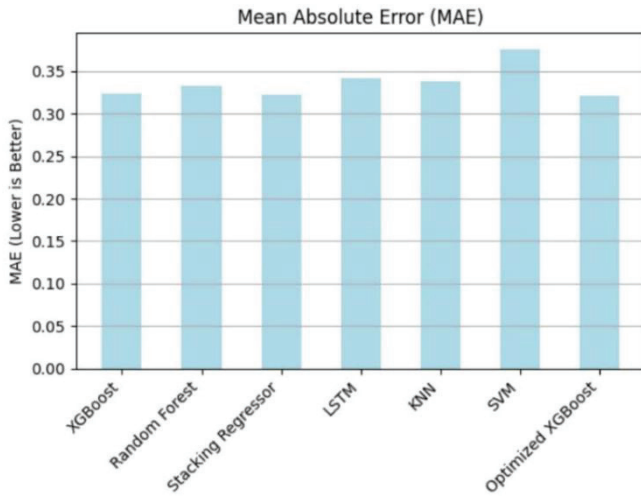


Fig. 9. Comparison of the mean absolute error among machine learning models.

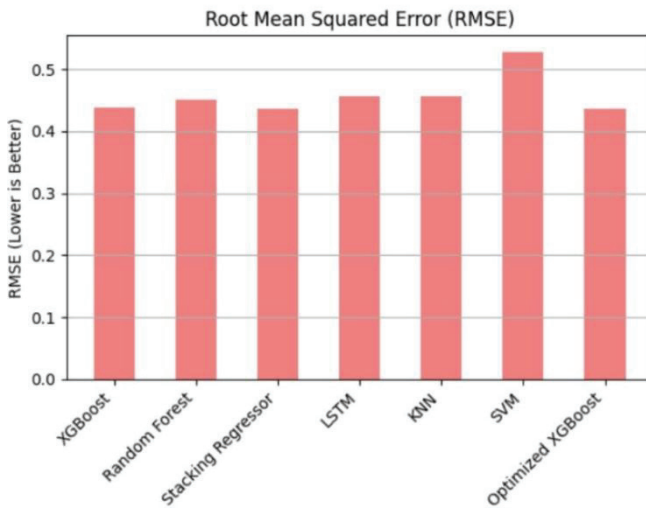


Fig. 10. Comparison of the root mean squared error among machine learning models.

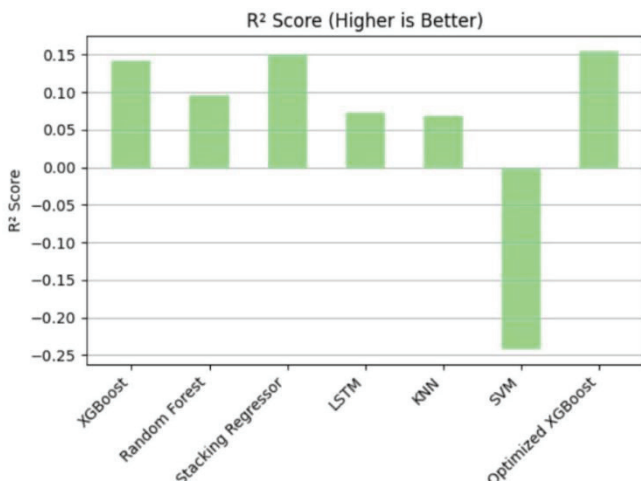


Fig. 11. Comparison of the R² Score among machine learning models.

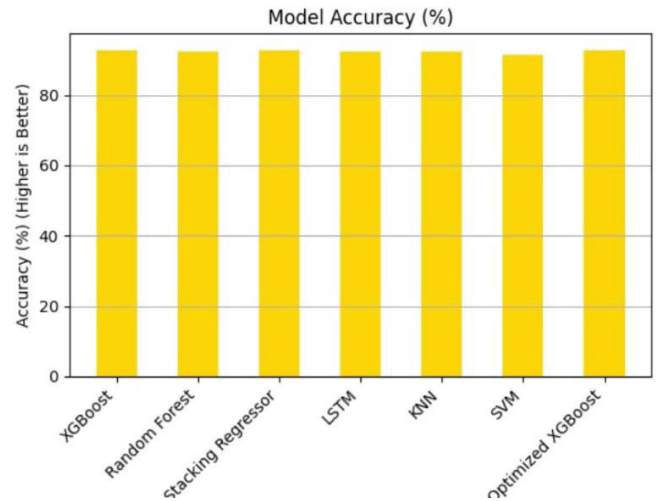


Fig. 12. Comparison of the Model Accuracy among machine learning models.

V. CONCLUSION

This research evaluates the best model for earthquake prediction, this study compares six machine learning algorithms: XGBoost, Random Forest, Stacking Regressor, LSTM, SVM, and KNN. The difficulty of predicting earthquake severity with few features was highlighted by the different levels of accuracy displayed by each model. The study offers an extensive review of the patterns of earthquakes across the Indian subcontinent from 2000 to 2024, highlighting high-risk regions with a majority of moderate-magnitude, shallow-depth earthquakes, such as the Andaman-Nicobar Islands and the foothills of the Himalayas.

Following hyperparameter adjustment, XGBoost outperformed the other models in terms of accuracy. Its predictive performance, stability, and generalization were improved by fine-tuning parameters like the learning rate, maximum depth, gamma, and regularization.

However, the future studies need to investigate additional variables that may improve the predictive accuracy, soil composition, aftershocks consequences, geological parameters, and real-time monitoring. In addition to, applying transfer learning in different earth-shaking areas could lead us a more omnipresent prediction model of earthquake. The actuality parameters of this study emphasize the real-world application of implementing real-time in early warning systems aiming to lower down casualties of populations living in the risk prone areas.

REFERENCES

- [1] H. B. G, C. G. Kumar S, P. U. R, and S. Anjum, "EARTHQUAKE PREDICTION MODEL USING MACHINE LEARNING," Jul. 2023. [Online]. Available: www.jetir.org
- [2] K. B. Y *et al.*, "Earthquake detection and early warning prediction using folium and Geopandas," *Cogent Eng.*, vol. 11, no. 1, 2024, doi: 10.1080/23311916.2024.2345301.
- [3] S. A. Kumar, A. Kumar2, A. Dhanraj, and A. Thakur, "Earthquake Prediction using Machine Learning," *Int. Res. J. Eng. Technol.*, 2021, [Online]. Available: www.irjet.net
- [4] F. Ahmed, S. Akter, S. M. M. Rahman, J. Bin Harez, A. Mubasira, and R. Khan, "Earthquake Magnitude Prediction Using Machine Learning Techniques," in *2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation, IATMSI 2024*, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/IATMSI60426.2024.10502770.

- [5] Kaladevi A C, G. M. L, and Iswarya K, "A Comparative Study on Earthquake Prediction using Machine Learning Algorithms."
- [6] Y. Shi, B. Zhang, S. Zhang, and H. Zhang, "On numerical earthquake prediction," *Earthq. Sci.*, vol. 27, no. 3, pp. 319–335, 2014, doi: 10.1007/s11589-014-0082-z.
- [7] N. B. Jarah, A. H. Hassin Alasadi, and K. M. Hashim, "Earthquake prediction technique: a comparative study," *IAES Int. J. Artif. Intell.*, vol. 12, no. 3, pp. 1026–1032, Sep. 2023, doi: 10.11591/ijai.v12.i3.pp1026-1032.
- [8] O. Dikmen, "Comparative Analysis of Machine Learning Models for Earthquake Prediction: A Case Study of Düzce, Türkiye," *Int. J. Innov. Res. Eng. Manag.*, vol. 11, no. 5, pp. 73–82, Oct. 2024, doi: 10.55524/ijirem.2024.11.5.10.
- [9] X. Wang *et al.*, "Small Earthquakes Can Help Predict Large Earthquakes: A Machine Learning Perspective," *Appl. Sci.*, vol. 13, no. 11, Jun. 2023, doi: 10.3390/app13116424.
- [10] B. Sadhukhan, S. Chakraborty, and S. Mukherjee, "Predicting the magnitude of an impending earthquake using deep learning techniques," *Earth Sci. Informatics*, vol. 16, no. 1, pp. 803–823, 2023, doi: 10.1007/s12145-022-00916-2.
- [11] N. B. Jarah, A. H. H. Alasadi, and K. M. Hashim, "A New Algorithm for Earthquake Prediction Using Machine Learning Methods," *J. Comput. Sci.*, vol. 20, no. 2, pp. 150–156, 2024, doi: 10.3844/jcssp.2024.150.156.
- [12] O. S. Atiyah, "Using Soft Voting Classifier of Machine Learning Algorithms for Earthquake Prediction," *Al-Kitab J. Pure Sci.*, vol. 9, no. 01, pp. 1–13, Dec. 2024, doi: 10.32441/kjps.09.01.p1.